

Title

Your name here

February 3, 2026

Exercise 1 [2]

Collaborators: Jared Contreras

Tools and resources used: On Exercise 1, I used ChatGPT to help me figure out how to [...]. On Choice Exercise 7, I watched a YouTube video to [...].

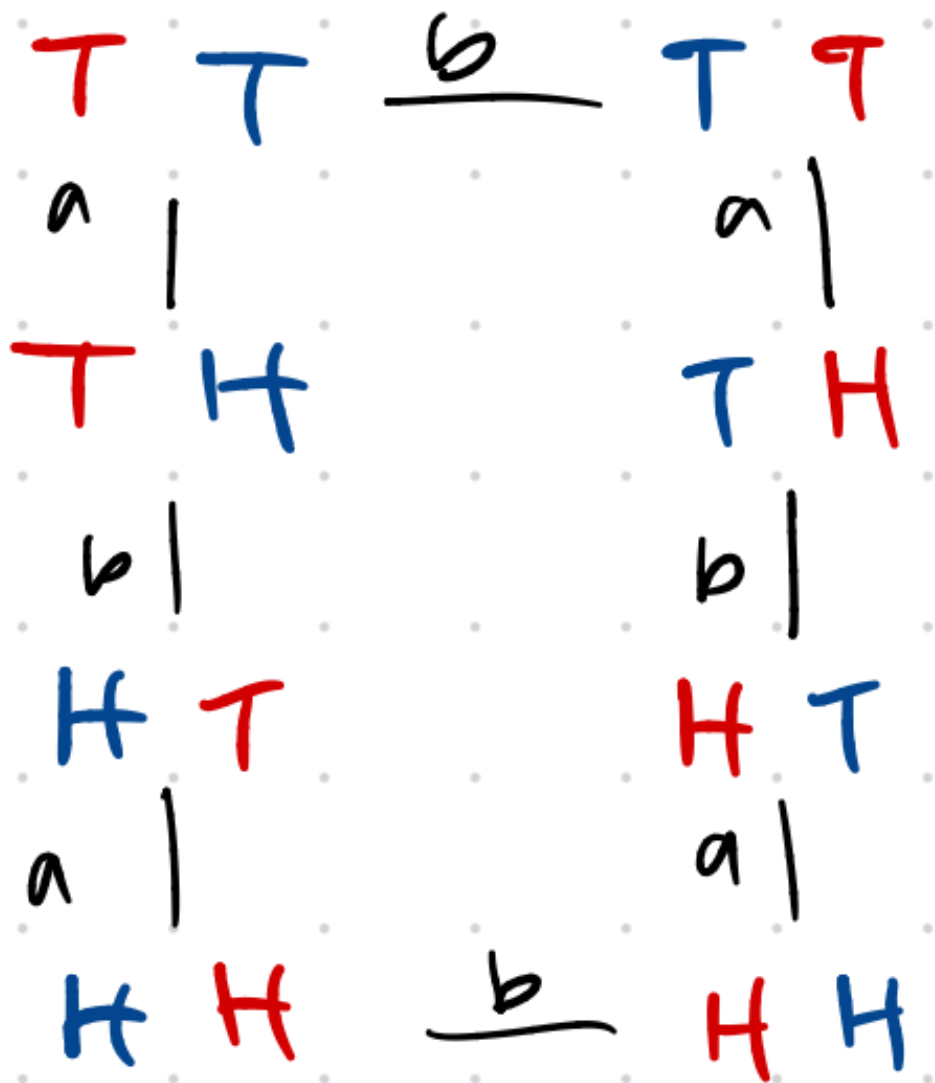
- Exercise 1 Points attempted: 2. Time spent: 5 minutes. Difficulty: 1/10. Self-assessed score: 2. Questions/Feedback: None.
- Exercise 2 Points attempted: 1. Time spent: 2 minutes. Difficulty: 1/10. Self-assessed score: 1. Questions/Feedback: None.
- Exercise 3 Points attempted: 6. Time spent: 30 minutes. Difficulty: 3/10. Self-assessed score: 6. Questions/Feedback: Did I do the problem correctly and as intended?

Exercise 2 [1]

Done! Posted an image of a marker with rotational symmetry.

Exercise 3 [6]

1. The coins have 8 symmetries. That is because there are 2^2 configurations without swapping places. When allowing swapping places, we get $2 \times 2^2 = 8$.
2. First, I chose the symmetry of flipping the *left*, denoted as f_L , coin over. Applying this symmetry twice gets us back to id. However, when we allow the swapping of places, denoted s , we can actually get to all states. We do not actually need an explicit 'flip right' symmetry, since it is actually equal to $f_L \circ s$.
3. In the diagram, let $a = f_L$ and $b = s$.



4. I used 2 generators.
5. This group is not commutative. This is because $f_L \circ s \neq s \circ f_L$. Swapping the coins then flipping the left is not the same as flipping then swapping.